10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 8.2 — Equal Chords, Equal Arcs and Equal Angles

Unit 8 | Solved Exercise | New Book (PECTAA)

Exercise 8.2 (Solved)

This exercise is all about the chain: equal chords → equal arcs → equal central angles. An arc of 3 cm means the arc has length 3 cm, and an angle of 70° at the centre means the arc in front of it measures 70°.

1. In the given figure, if mPQ = mQR = mRS = mST and m∠POT = 70°, then find the value of x.

The four chords PQ, QR, RS and ST are equal

Equal chords of a circle subtend equal angles at the centre

Let each of these angles be θ

∠POQ = ∠QOR = ∠ROS = ∠SOT = θ = 70°

x = ∠SOR + ∠SOT = 70° + 70°

x = 140°

Answer: x = 140°

2. In the given figure, find the values of x and y.

In the figure the chords AB, BC and CD are marked equal, ∠BOD = 134° and the arc AB is 3 cm long.

∠BOD = ∠BOC + ∠COD = 134°

Equal chords BC and CD subtend equal angles at the centre

$\angle BOC = \angle COD = \frac{134°}{2} = 67°$

x = ∠COD = 67°

Equal chords have equal arcs: arc CD = arc BC = arc AB

y = arc CD = 3 cm

Answer: x = 67°, y = 3 cm

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3. Find the values of x and y in the given figure, such that mAB = mBD.

In the figure ∠AOB = 80°, ∠BOC = x, ∠COD = 52°, arc AB = 6 cm, arc BC = 2 cm and arc CD = y cm.

Arc AB = arc BD

80° = x + 52°

x = 80° − 52°

x = 28°

Equal arcs have equal lengths, so arc BD = arc AB = 6 cm

arc BC + arc CD = 6 cm

2 + y = 6

y = 4 cm

Answer: x = 28°, y = 4 cm

4. Two congruent arcs in a circular track subtend angles of 60° each at the centre. If the length of one chord of the circle is 10 metres, what is the length of the other chord?

Congruent arcs subtend equal central angles

Both arcs subtend 60° at the centre

Equal central angles give equal chords

Answer: Length of the other chord = 10 m

5. In a circular fountain, two water jets spray water along arcs of equal length. If one jet sprays between P and Q and the chord PQ is 12 metres, what is the chord RS of the second jet along an arc congruent to arc PQ?

The two arcs are congruent, so they subtend equal angles at the centre

Equal arcs of a circle have equal chords

chord PQ = 12 m $\implies$ chord RS = 12 m

Answer: Chord RS = 12 m

6. In a circular park, two walkways AB and CD are both straight paths (chords) of length 14 metres. What can be said about the minor arcs subtended by these walkways?

AB = CD = 14 m

Equal chords of a circle subtend equal (congruent) arcs

Answer: The two minor arcs are congruent, arc AB ≅ arc CD

7. In two congruent circular clocks, the minute hand points from the centre to the 3 on both. A decorative string connects 3 to 9 on both clocks. Are the arcs from 3 to 9 on both clocks congruent?

The clocks are congruent, so their radii are equal

On each clock 3 to 9 is a semi-circle, that is an arc of 180°

Equal radii and equal central angles give congruent arcs

Answer: Yes, the two arcs are congruent